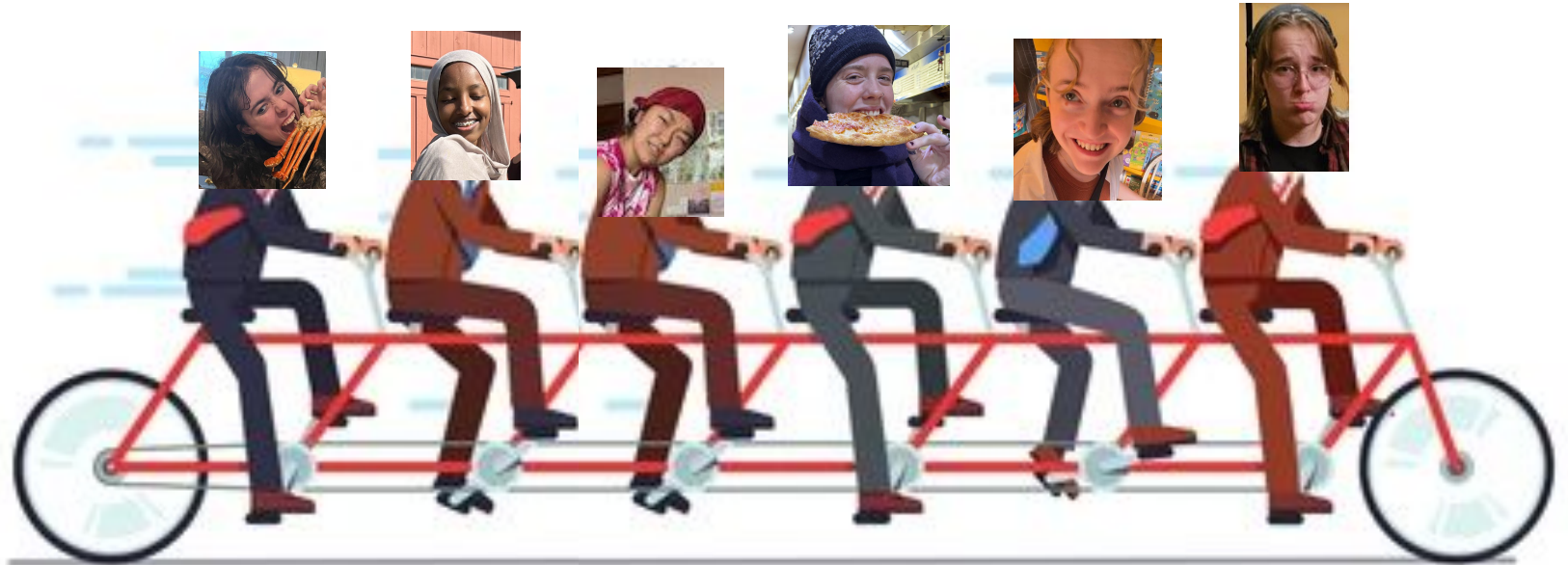
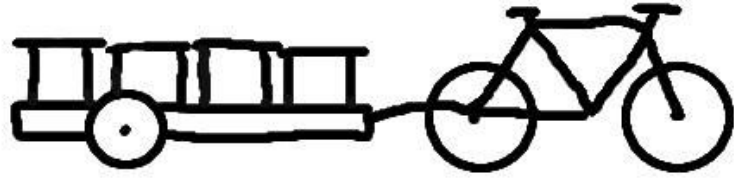


Final Demo Day

By Bike Buddies



What is Pedal People?



- Pedal people is a worker-owner co-op based in Northampton
 - Trash, compost, and recycling pickup
 - Hauling services
 - All by **bike**!



Pedal People Website



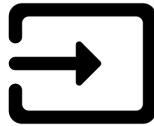
Project Goals

Calculate carbon emissions saved from...



Create an emissions calculator....

Pickup data (gal)

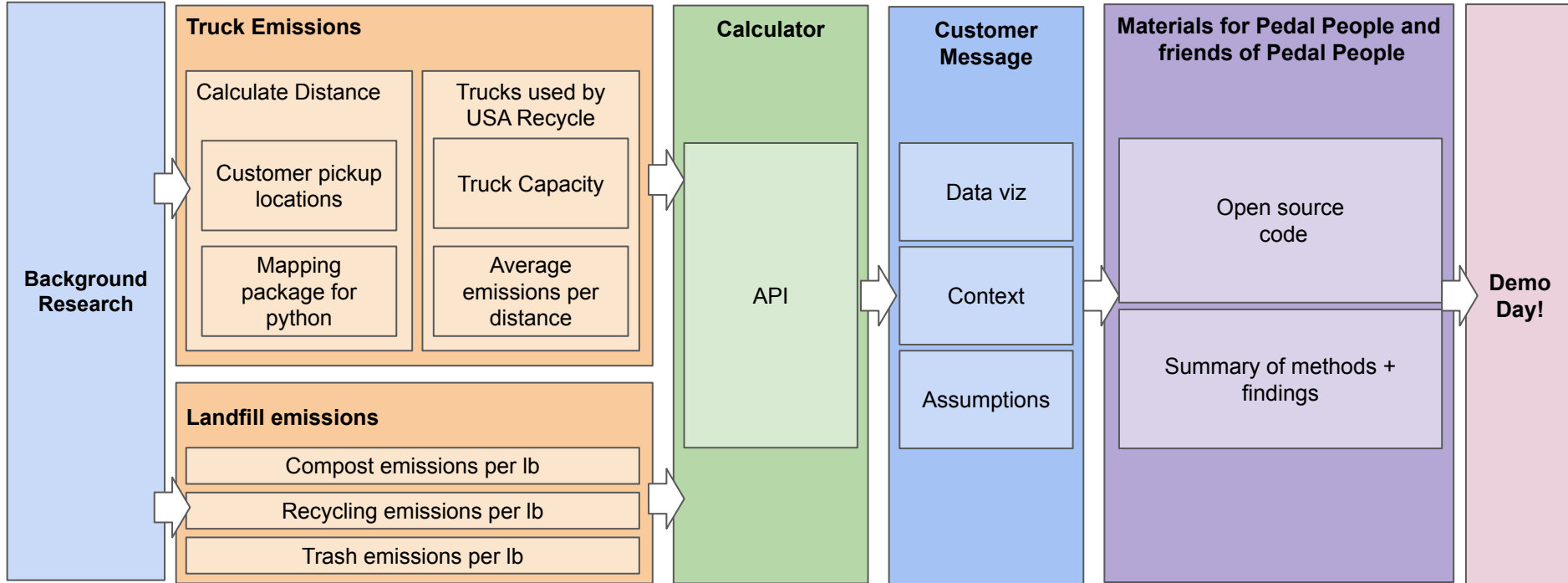


CO₂ emissions (lbs)

Sprint Diagram

Future Directions →

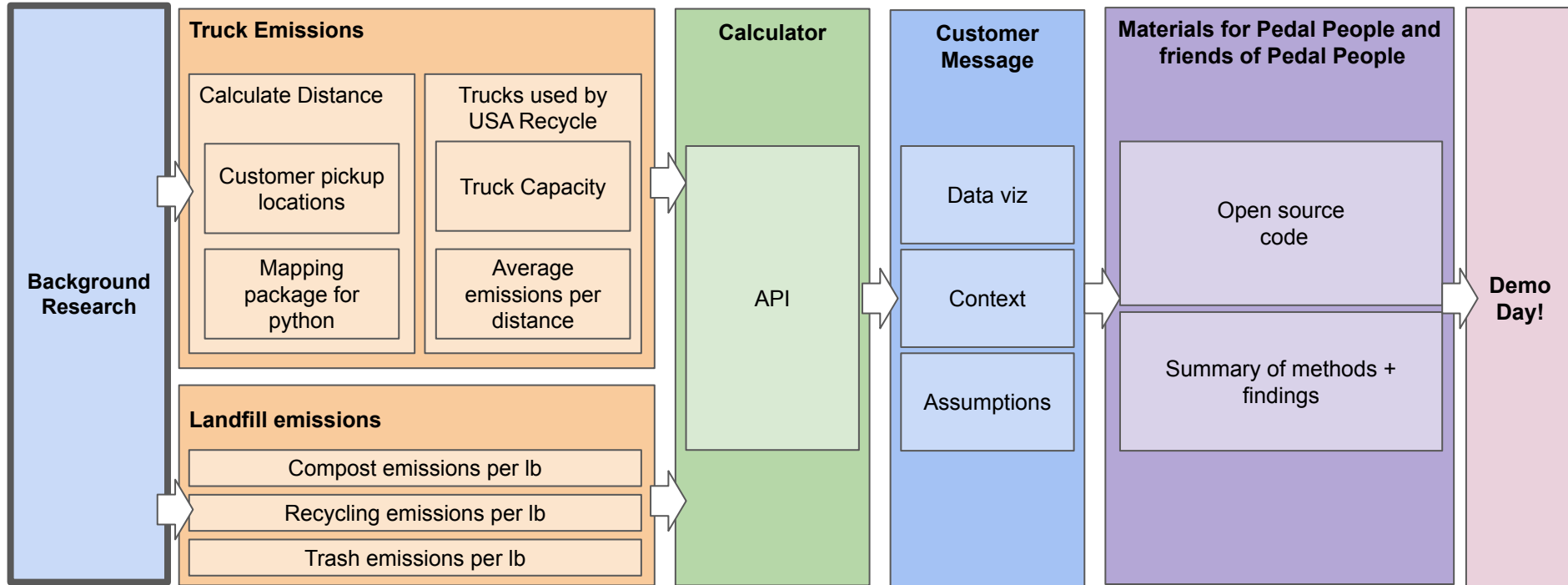
Making code compatible to
datasets with new customers



Sprint 1

Future Directions →

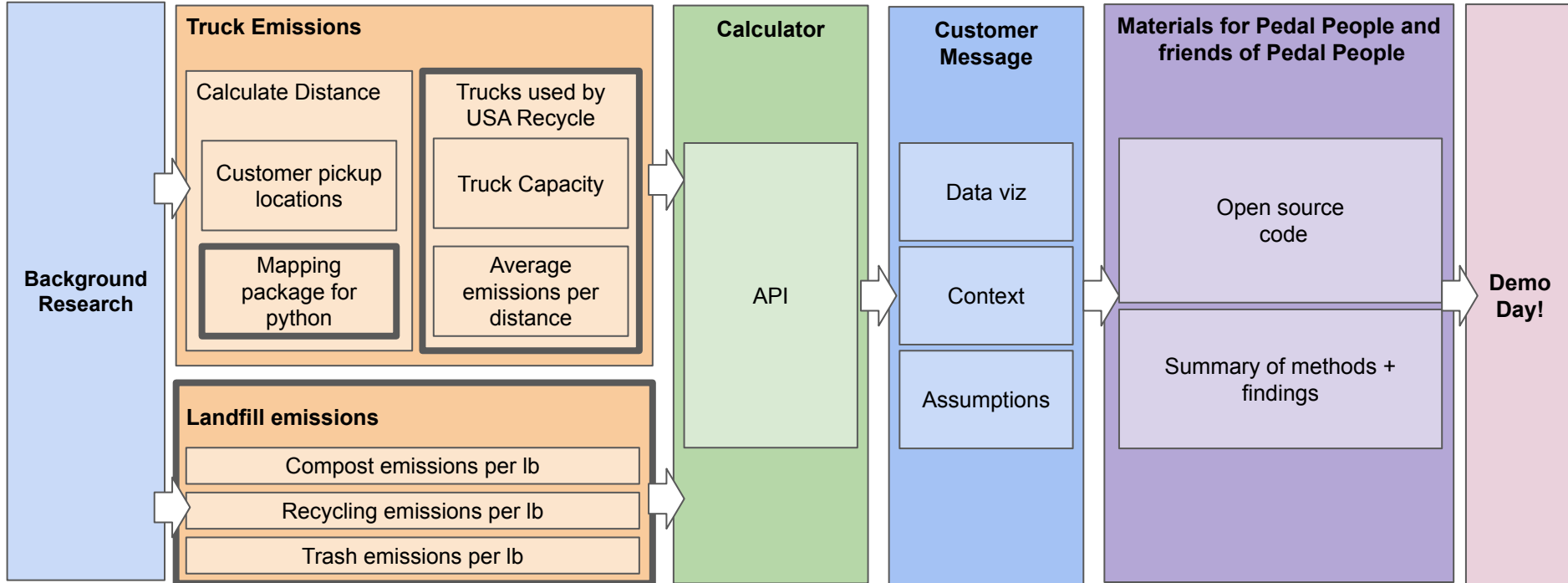
Making code compatible to
datasets with new customers



Sprint 2

Future Directions →

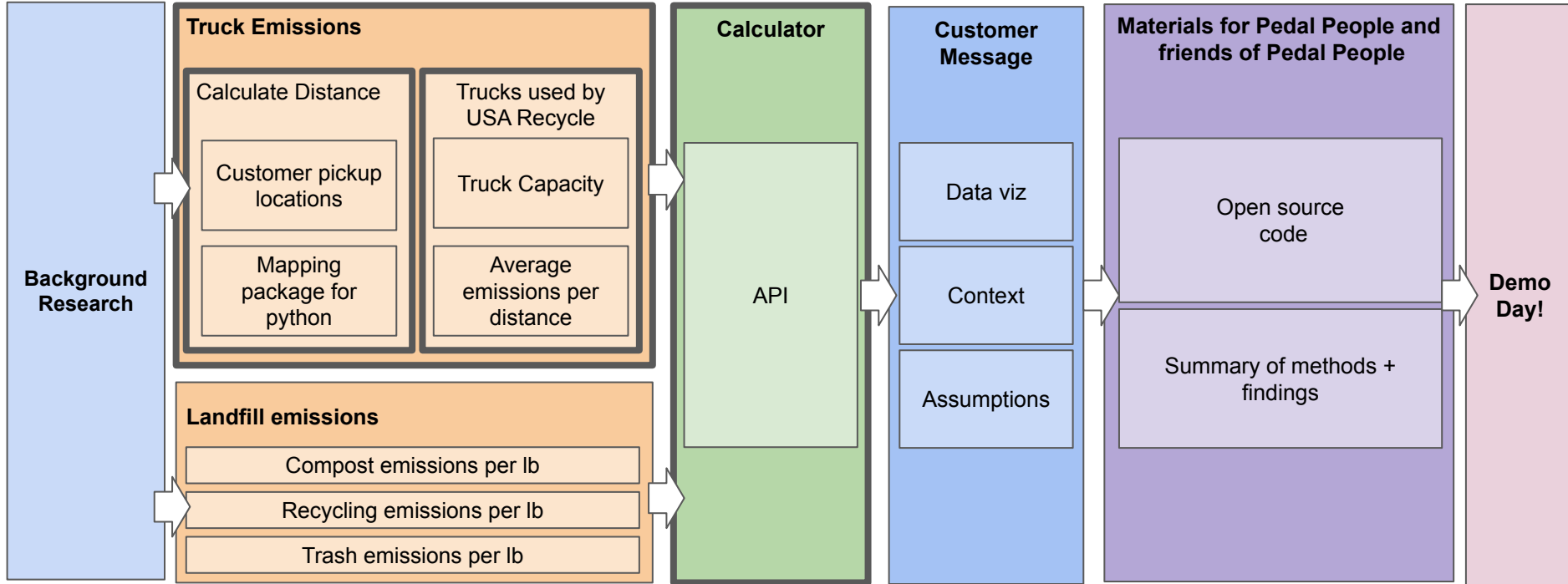
Making code compatible to datasets with new customers



Sprint 3

Future Directions →

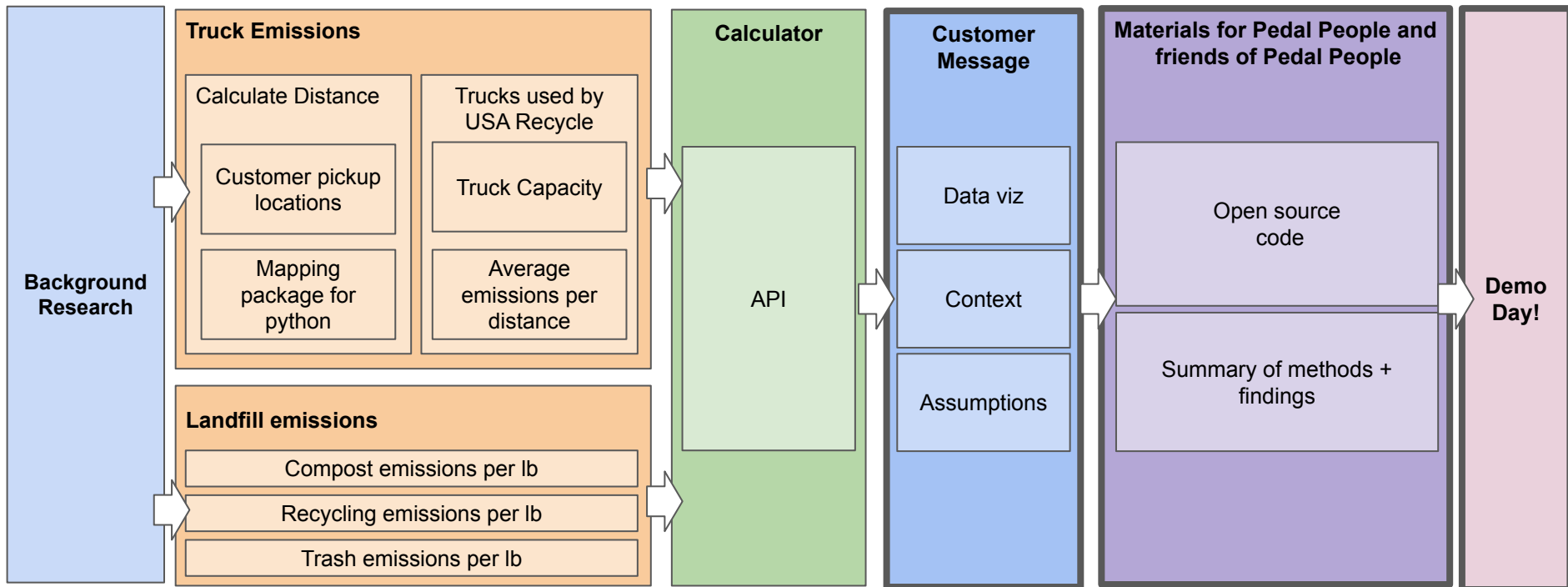
Making code compatible to datasets with new customers



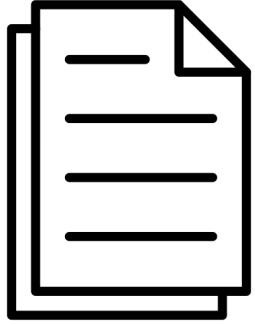
Sprint 4

Future Directions →

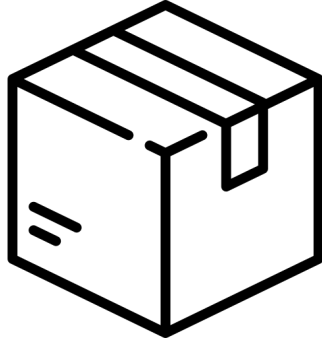
Making code compatible to datasets with new customers



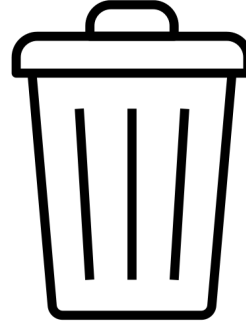
Landfill Emissions



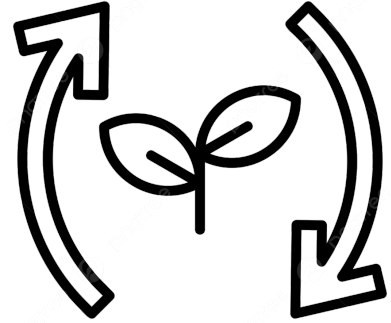
Paper



Containers



Trash

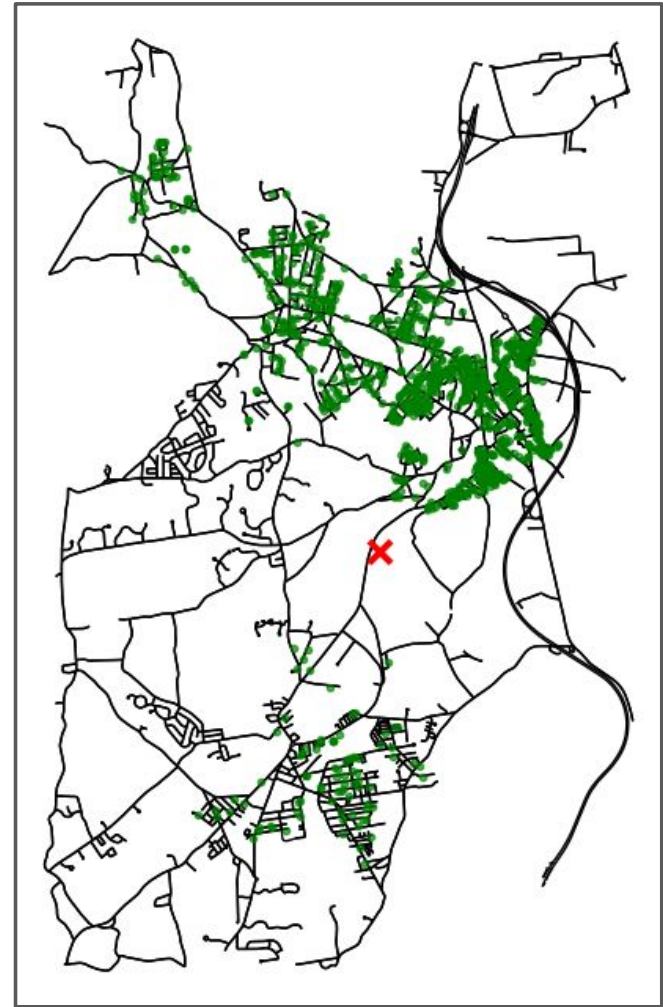
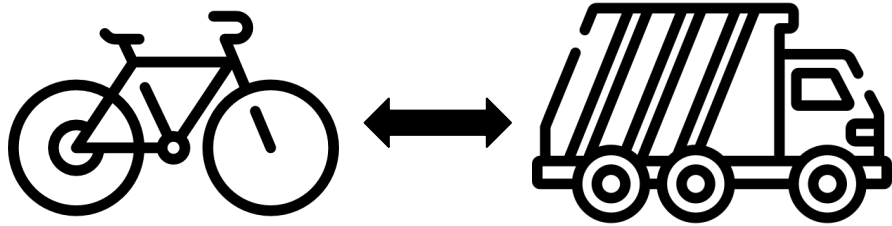


Compost

Used volume-to-weight conversion factors from the EPA to convert our input in gallons to weight (either lbs or cubic yards) so that we could use an emissions factor that represents the lbs of CO₂ per gallon of a particular material

Transportation Emissions

- How much emissions are saved by picking up with bike instead of truck?
- We need:
 - Truck information
 - Route



Automated Side Loader Trucks

- Used by local competitor for residential waste pickups
- Runs on compressed natural gas
 - Emits ~ 39.22 lbs/mile
- Capacity
 - 28 cubic yards



Truck Route

- Created using Dijkstra's algorithm and the TSP algorithm
- Total route distance
 - Noho route: 63.5 miles
 - Easthampton route: 21.25 miles
- Routes start and end at Valley Recycling
- Combining all stops, the material in Northampton would be over capacity
 - The truck will run twice: once for trash, once for recycling



Truck Emissions Calculation

$$\text{Transportation} = \frac{\beta_{Truck}(\text{Mileage})}{N} * 2$$

$$\beta_{Truck} = 39.22 \text{ lbs CO}_2/\text{mile}$$

Distance = 63.5 miles in Noho, 21.25 miles for Easthampton

N = Number of pickups total in either Noho or Easthampton

5.3 lbs CO₂ per person, per pickup (trash & recycling; Noho)

19.4 lbs CO₂ per person, per pickup (trash & recycling; Easthampton)

Python - Emissions Calculator



- Built a series of functions calculating the amount of emissions produced per unit of waste
 - Separate for type of waste (landfill, compost, containers, paper)
 - Convert waste from gallons to lbs (standard unit for emissions measurement)
- Also built a function using the mileage derived from step to calculate transportation emissions (shown in previous equation slide!)
- Each function includes a unique emissions constant (i.e., emissions per unit of food compost, per mile of driving, etc.)

Python - Emissions Calculator



Example: One of our functions

```
def trash_gallons_to_co2(gallons: float) -> float:
    """
    take in gallons of trash and return CO2 emissions
    calculations (from hannah)
    - 1 gal = 0.004951 cubic yards
    - 0.004951 cubic yards * 275 lbs/cubic yard = 1.361525 lbs MSW/gal
    - 1.361525 lbs * (1 ton / 2000 lbs) = 0.0006807625 tons MSW/gal
    - 902 kg CO2e / short ton MSW (EPA) = 0.994285 tons CO2e/ton MSW
    - 0.994285 * 0.0006807625 = 0.00067687194 tons CO2e/gal
    - 1 ton = 2000 lbs --> ~1.49 lbs CO2e/gal
    """

    TONS_MSW_PER_GALLON = 0.0006807625
    TONS_CO2_PER_TON_MSW = 0.994285
    LBS_PER_TON = 2000

    tons_co2 = gallons * TONS_MSW_PER_GALLON * TONS_CO2_PER_TON_MSW
    lbs_co2 = tons_co2 * LBS_PER_TON

    return round(lbs_co2, 4)

trash_gallons_to_co2(tiny_df['trash'].iloc[0])
```

Python - Emissions Calculator



Ultimately, we sum together values and get monthly emissions per customer:

```
Waste = tiny_df['frequency']*(trash + paper + containers + compost)

Transportation = tiny_df['frequency']*(individual_transport(tiny_df['city'].iloc[0]))

Without_PP = Waste + Transportation

With_PP = Waste
```


Python - Emissions Calculator



We can also derive a “worst case” scenario if all items were put in the landfill

```
trash_trash = trash_gallons_to_co2(tiny_df['trash'].iloc[0])
paper_trash = trash_gallons_to_co2(tiny_df['paper'].iloc[0])
containers_trash = trash_gallons_to_co2(tiny_df['containers'].iloc[0])
compost_trash = trash_gallons_to_co2(tiny_df['compost'].iloc[0])

Waste_bad = tiny_df['frequency']*(trash_trash + paper_trash + containers_trash + compost_trash)

Without_PP_trash = Waste_bad + Transportation

With_PP_trash = Waste_bad
```

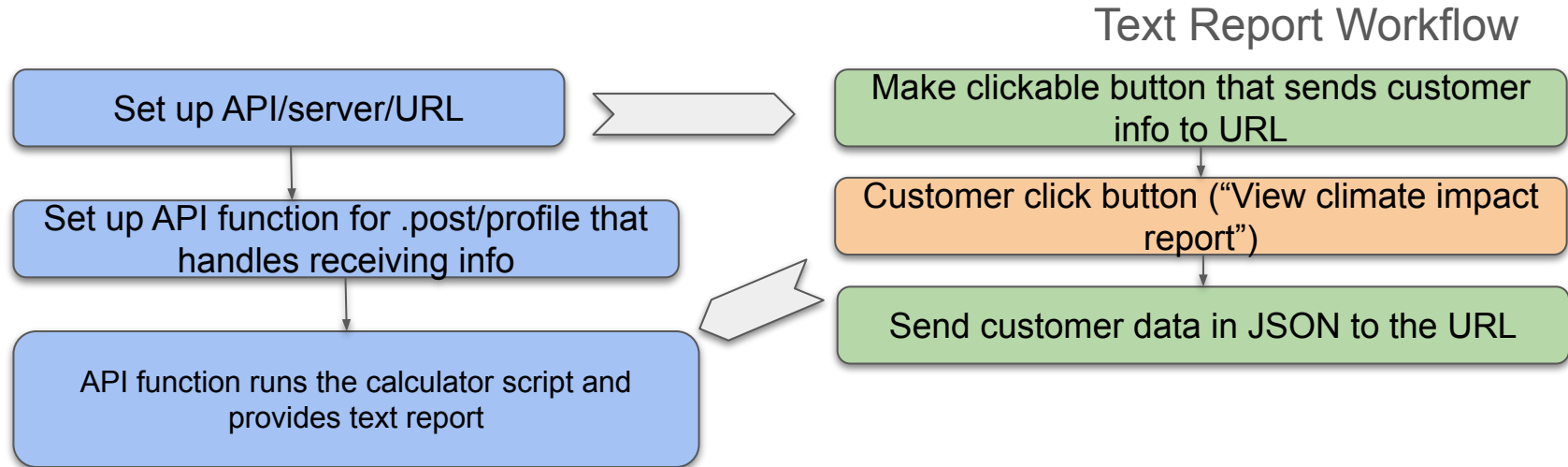
How do we put this on the Pedal People website?

Using an API!

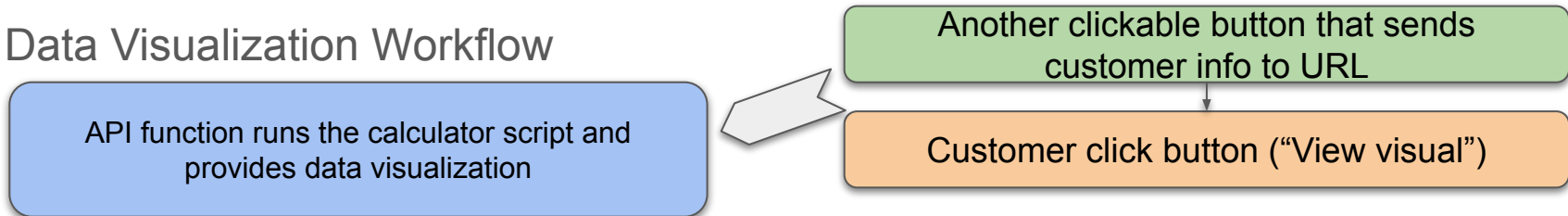


- We have a script in Python, a language that we are familiar with
- Pedal People's website is running using PHP, a language that we are not so familiar with
- An API (Application Programming Interface) allows our script to communicate with the Pedal People website without anyone having to run something manually every time a customer wants to view their climate reduction report

What does our API do/How does it work?



Data Visualization Workflow



Using API to run script

- <http://127.0.0.1:8000/docs#/>



Current customer output



recycling, trash & compost

- [Schedule a one-time pickup](#)
- [Sign up for a new regular pickup](#)
- [View tiered pricing options](#)

(active)

- Plan: **biweekly trash and recycling (30 gallons of trash)**
- Price tier: **2** ([learn more or change](#))
- Monthly price: **\$34.00 - \$-34.00 (two-family pickup \$4.50; barter -\$38.50) = \$0.00**
- Pickup day: every other Saturday (next pickup Sat, May 2)
- 3-month average: 18 gal. trash, 10 gal. paper, 14 gal. containers, 0 gal. compost
- Compost buckets on loan: 0; recycling totes on loan: 0; housing units: 2; [city bags](#): no
- Resident contacts: None ([edit](#))
- Automated reminder sent the night before the pickup: Disabled ([edit](#))

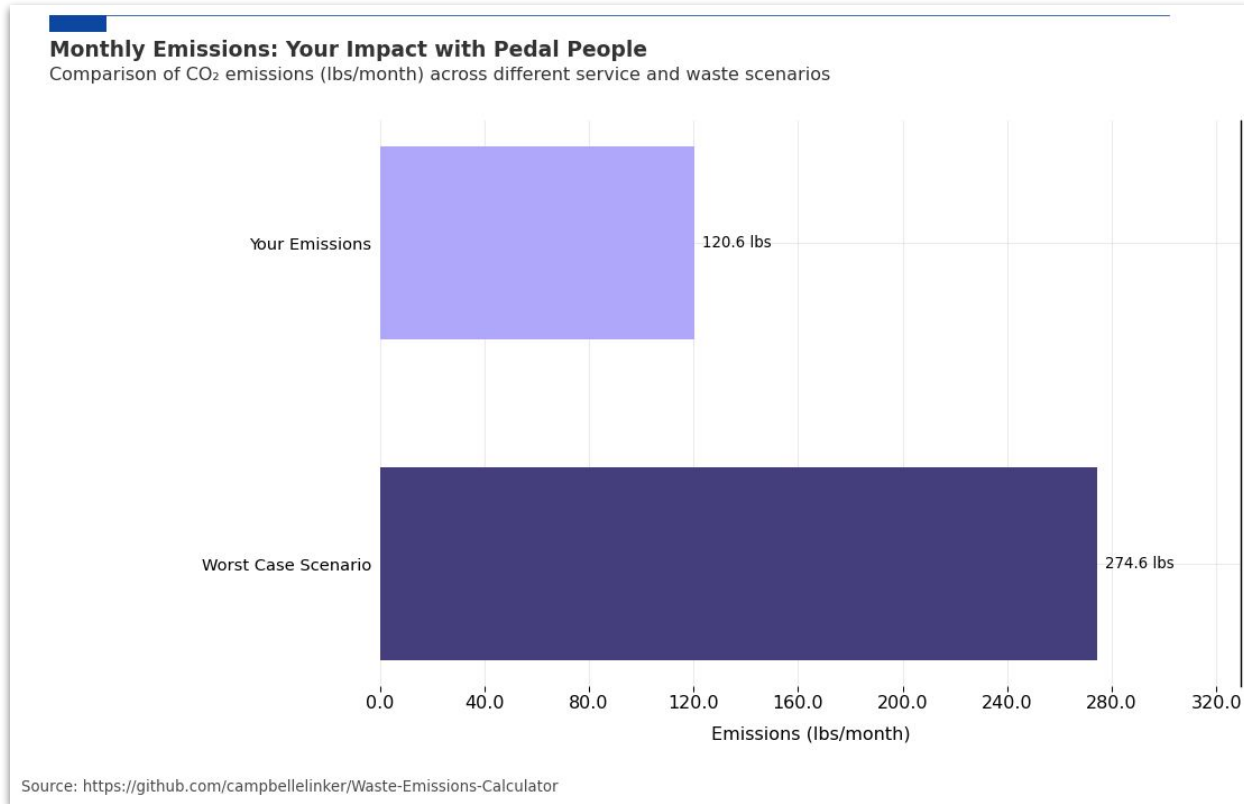
Carbon reduction information: Thank you for using Pedal People! We have compiled a report to share with you the positive impact you had on the environment by taking proper care of your waste and using Pedal People to haul your trash, recycling, and compost. This month, your use of Pedal People's services significantly reduced your carbon footprint, saving approximately 136.53 lbs of CO2 emissions by properly disposing of your recycling, trash, and compost through Pedal People. This is compared to how many pounds of CO2 your household would emit in one month if you threw all recycling, trash, and compost away for a truck to take to a landfill. By using Pedal People to haul your trash instead of a traditional trash truck, you saved 10.59 lbs of CO2 emissions this month. This is equivalent to: - Driving from City X to City Y - Approximate amount of CO2 consumed by X fully-grown trees in one year, on average



contact

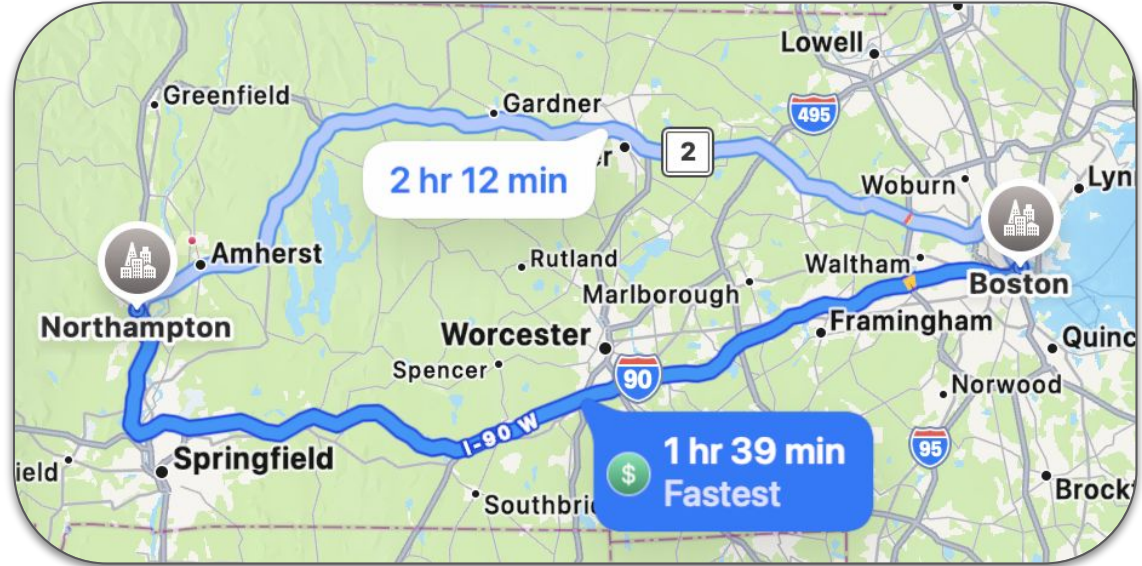
413-586-8591 (call or text) | [Email us](#)
PO Box 415, Northampton, MA 01061

NEW Customer output - data viz

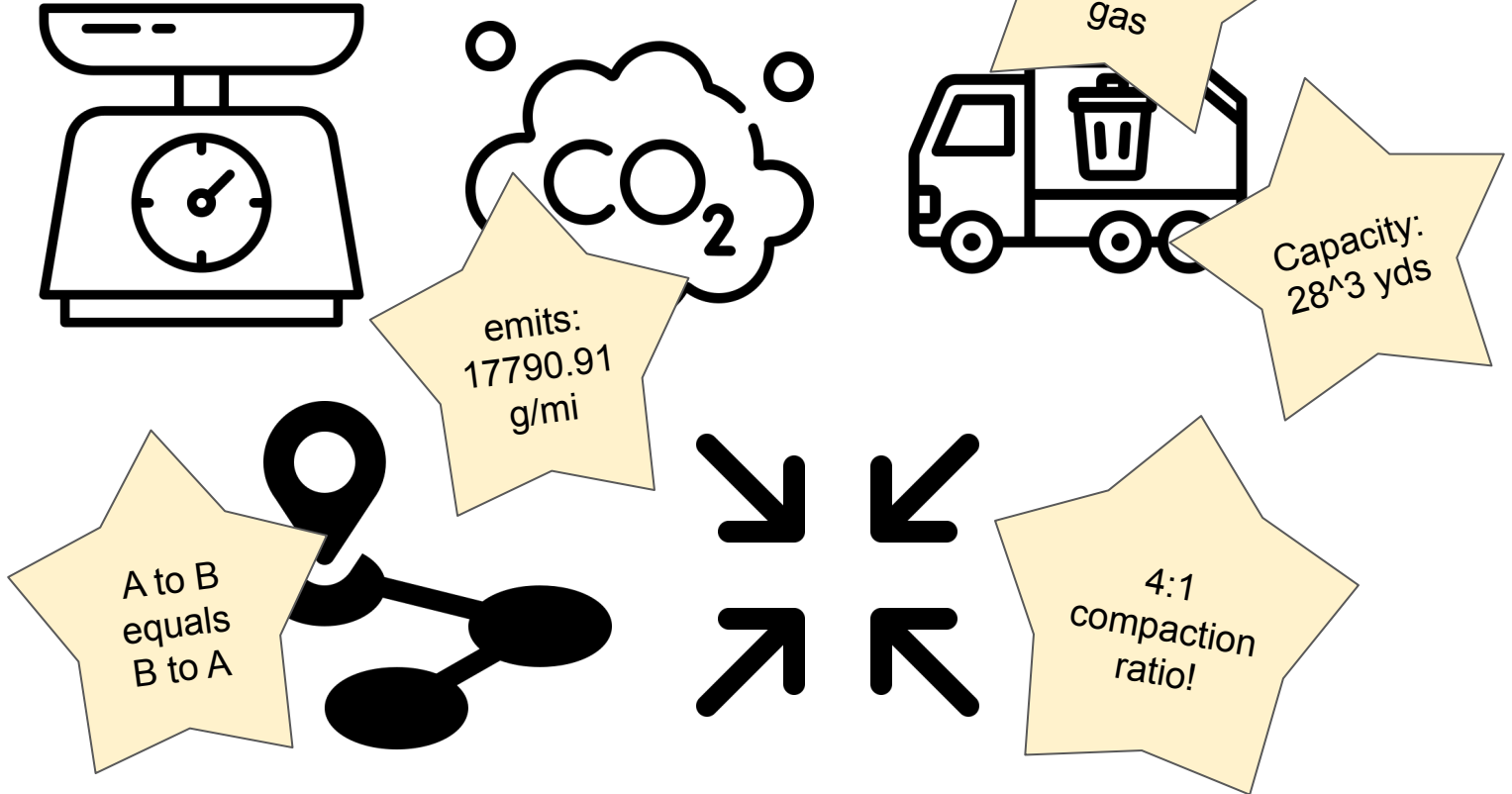


NEW Customer output - contextualization

- Compares emissions saved to the emissions for driving from Northampton to relevant cities (Amherst, Springfield, Boston, New York, and Philadelphia)
- “This is equivalent to driving from Northampton to Boston, MA, 3 times!”



Our Assumptions



Future Directions

Sharing our work online: documenting code and assumptions

- <https://github.com/campbellelinker/Waste-Emissions-Calculator>
- Helpful for anyone attempting to solve similar problems in this niche area of research
- Adding to this, in addition to sponsor-specific documentation





Any questions?

